Asgard@SpacePole XV (2025-26)

Students design, develop, test and fly their science/technology experiment to the edge of space with a stratospheric balloon

Application form for secondary schools

To be completed and returned by email to:
eds@sint-pieterscollege.be
(St-Pieterscollege Jette, Brussels)
before november 11th, 2025

You apply as:
Beginners team

School name and address:
The English College in Prague
Sousedíkova 1044/8, 190 00 Praha 9,
Czech Republic

Teacher email:
Jiri Dobes, jdobes.cz@gmail.com (private individual, not a teacher)

Teacher mobile phone:
+420 734 518 454

Age and study orientation of the students:
English high school, age:13-17

Experiment name or title:
Stratospheric Measurements of Cosmic-Ray-Induced Single-Event Upsets in Static Random-Access Memory

Objective of the experiment (this is the so-called 'research question'):
The objective of this experiment is to investigate the rate and character of cosmic ray-induced bit flips, also known as single-event upsets, in memory devices in the stratosphere (20-35km above sea level) and to compare the observed rate of single-event upsets with existing theoretical expectations.

Relevance of the experiment (motivation of the 'research question'):
This experiment is highly relevant largely because of the significant data gap in stratospheric single-event upset (SEU) measurements. Though there is some existing data, it largely comes from accelerator beam tests, which are limited in energy range and ion species compared to the mixed flux found in the stratosphere. There are very few in situ experiments that concern the rate and character of SEUs in the stratosphere. Furthermore, the data obtained in this experiment could prove useful in

the development of commercial off-the-shelf parts for cube-sats and high-altitude avionics. This experiment would also be relevant for reliability assessments of near-space systems. It addresses a genuine gap in the scientific understanding and is highly relevant to its particular field. As systems progressively become more and more digital, mitigating the effects of bit flips will become more important.

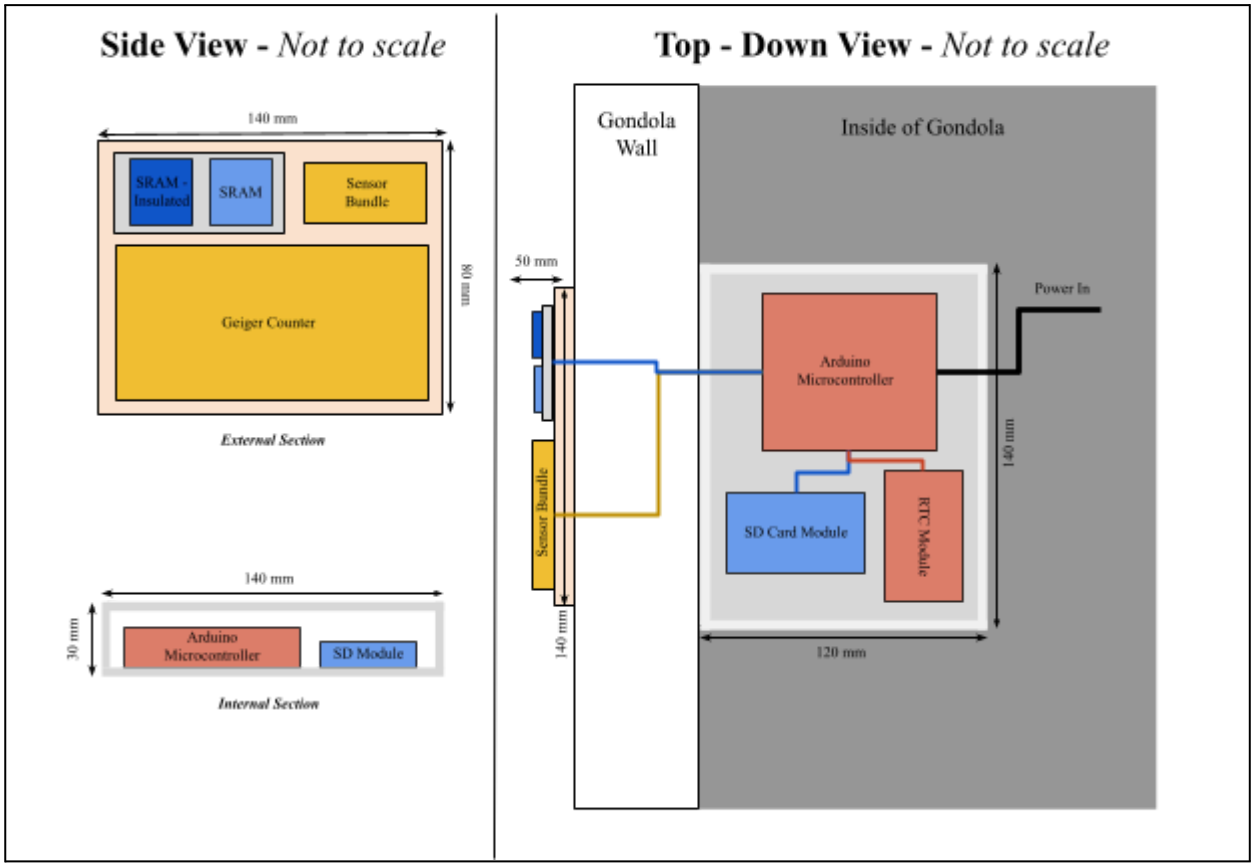
Short description of the experiment:

Our experiment has two sections: an external one mounted outside the gondola and an internal one mounted inside. Both sections are interconnected by wires leading from the gondola outside.

The first components of the external section are two SRAM microchips. One SRAM chip is enclosed within a local metallic shield to serve as a relative control, and the other is exposed directly to the radiation environment. These are attached to a mini breadboard. The microchips are continuously written with known data patterns, and periodically checked for bit flips. The second component is a Geiger counter module. The Geiger counter provides an approximate indicator of radiation activity, which may be compared qualitatively with observed SEU rates. The third component is a joint air pressure, temperature, and humidity sensor, which records these values and allows us to separate the effects of these factors from the isolated effects of cosmic radiation during interpretation.

The internal section is made up of components enclosed within a 3D printed, thermally insulated casing to protect against temperature extremes. The first component of the internal section is an Arduino microcontroller, which controls and gathers readings from all other components of our experiment. The second component is an RTC module, which timestamps all recordings from the experiment with the current time, preventing discontinuity if the Arduino resets. The third component is an SD card module, which is thoroughly insulated from cosmic radiation, and stores the readings of all other components of our experiment for later interpretation.

Scheme or drawing of the experiment:



Estimated mass of the experiment (max 200g, but small is beautiful!):
190g

Estimated dimensions of the experiment in mm (L x W x H):
Internal section: 140mm x 120mm x 30mm,
External section: 140mm x 80mm x 30mm

Do you wish to use the shared Asgard batteries (max 200mA@5VDC)? YES

If YES: estimated current draw @5VDC in mA:
Yes, 105mA up to 180mA during SD writes

Do you plan on taking your experiment to Sint-Pieterscollege for pre-integration testing in the space-simulation chamber? (If YES: Contact Erik for interfacing and test details.

Yes

REMARK: If your experiment cannot possibly be realised within the mass (and current) constraints you must <u>apply</u> for a waiver at this time. This requires written and argumented motivation! Waiver applications are considered during

selection. Asgard organisers decide on whether or not to grant waivers. Their decisions cannot be appealed.

Other characteristics (if relevant):
Part of the experiment needs to be mounted outside the gondola to ensure proper functioning of the apparatus, whereas other components need to stay inside for insulation.

If selected, do you grant Asgard organisers and Asgard partners the right to publish pictures and video material showing yourself and/or your students whilst engaging in all Asgard-related activities, for the purpose of promoting the Asgard programme or the activities of said partners?

X Yes

No

ATTENTION: In order to maximize the educational return for ALL students we insist that all teams take part in all campaign-related activities, INCLUDING the friday morning 'first results' presentations. Failure to comply may result in your experiment not being flown.
A team consists of 1 or 2 teachers and up to 5 students.

We are providing free of charge:

- the 3 days program in Brussels (22-24/4/2026)
- sandwiches on each of the 3 days

We do NOT provide:

- Night accommodation
- Transport or transport reimbursement (All locations are easily accessible by Brussels public transport)
- Dinners
- Hardware building/testing materials for your experiment

More info
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